第11讲. 含参定积分（续）

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作业：P332. 1(3). (4). 2. 3. 4(1). (4).

复习：① 设 $f(x,u) \in C(D)$. 其中 $D=[a,b]\times[\alpha,\beta]$. 则 $\varphi(u)=\int_a^b f(x,u)dx \in C[\alpha,\beta]$.

② 设 $f(x,u)$, $f_u'(x,u) \in C(D)$. $D=[a,b]\times[\alpha,\beta]$.

则 $\varphi(u)=\int_a^b f(x,u)\,dx$ 在 $[\alpha,\beta]$ 上可导. 且 $\varphi'(u)=\int_a^b f_u'(x,u)\,dx = \int_a^b \frac{\partial f(x,u)}{\partial u}dx$

$$\varphi'(u) = \frac{d\left(\int_a^b f(x,u)dx\right)}{du}$$

$$\lim_{u\to u_0}\varphi(u)=\varphi(u_0)=\int_a^b f(x,u_0)\,dx$$

$$\lim_{u\to u_0}\int_a^b f(x,u)\,dx = \int_a^b\left(\lim_{u\to u_0} f(x,u)\right)dx$$

定理1. 设 $f(x,u)$, $f_u'(x,u)\in C(D)$. $D=[a,b]\times[\alpha,\beta]$.

$a(u)$, $b(u)$ 在 $[\alpha,\beta]$ 上可导 且 $\forall u\in[\alpha,\beta]$. $a(u)$, $b(u)\in[a,b]$.

则 $\varphi(u)=\int_{a(u)}^{b(u)} f(x,u)\,dx$ 在 $[\alpha,\beta]$ 上可导. 且

$$\varphi'(u)=\int_{a(u)}^{b(u)} f_u'(x,u)dx + b'(u)f(b(u),u) - a'(u)f(a(u),u)$$

证：$\forall u\in[\alpha,\beta]$. 令 $y=a(u)$, $z=b(u)$.

$$F(u,y,z)=\int_y^z f(x,u)dx = -\int_z^y f(x,u)dx$$

$$\varphi(u)=F(u,a(u),b(u))$$

$$F_u'=\int_y^z f_u'(x,u)\,dx \in C(D)$$

$$F_y'=-f(y,u) \in C(D)$$

$$F_z'=f(z,u)\in C(D)$$

$$\varphi'(u)=F_u'+F_y'\cdot a'(u)+F_z'\,b'(u)$$

$$=\int_{a(u)}^{b(u)} f_u'(x,u)\,dx - f(a(u),u)\,a'(u) + f(b(u),u)\,b'(u).$$

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解: $F(x)=x\int_0^x f(u)\,du+\int_0^x uf(u)\,du$

$F'(x)=\int_0^x f(u)\,du+2xf(x)$

$F''(x)=3f(x)+2xf'(x)$.

引理1. 设 $f(x,y)\in C[a,b]\times[\alpha,\beta]$. $\forall x\in[a,b]$. $\forall t\in[\alpha,\beta]$.

令 $g(x,t)=\int_\alpha^t f(x,y)\,dy$ 则

① $g(x,t)\in C(D)$. $D=[a,b]\times[\alpha,\beta]$.

② $\forall x\in[a,b]$. $\alpha(x),\beta(x)\in[\alpha,\beta]$ 若 $\alpha(x),\beta(x)\in C[a,b]$. 则

$$F(x)=\int_{\alpha(x)}^{\beta(x)}f(x,y)\,dy\in C[a,b].\qquad \lim_{x\to x_0}F(x)=\int_{\alpha(x_0)}^{\beta(x_0)}f(x_0,y)\,dy$$

分析: $\forall (x_0,t_0)\in D$. $\lim\limits_{(x,t)\to(x_0,t_0)}g(x,t)=g(x_0,t_0)$

即证 $\forall\varepsilon>0$. $\exists\delta>0$. ? $\forall (x,t)$. $\sqrt{(x-x_0)^2+(t-t_0)^2}<\delta$

$$|g(x,t)-g(x_0,t_0)|<\varepsilon.$$

$$\left|\int_\alpha^t f(x,y)\,dy-\int_\alpha^{t_0}f(x_0,y)\,dy\right|$$

$$\le\int_\alpha^{t_0}|f(x,y)-f(x_0,y)|\,dy+\left|\int_{t_0}^t|f(x,y)|\,dy\right|\le M|t-t_0|$$

$\exists\delta_1>0$. s.t. $\forall x$. $|x-x_0|<\delta_1$. $\forall y\in[\alpha,\beta]$.

$$|f(x,y)-f(x_0,y)|<\frac{\varepsilon}{2(\beta-\alpha)}$$

$\because M>0$. s.t. $\forall(x,y)\in D$. $|f(x,y)|\le M$.

$\exists\delta_2>0$. s.t. $|t-t_0|<\delta_2<\frac{\varepsilon}{2M}$

$\delta=\min\{\delta_1,\delta_2\}$. 当 $\sqrt{(x-x_0)^2+(t-t_0)^2}<\delta$.

$\delta = \min\{\delta_1, \delta_2\}$. $\Rightarrow$ $\sqrt{(x-x_0)^2+(t-t_0)^2} < \delta$.

$|x-x_0| < \delta$. $|t-t_0| < \delta$.

$< \varepsilon$.

② $F(x) = \int_{\alpha(x)}^{\beta(x)} f(x,y)\,dy$

$$= \int_{\alpha}^{\beta(x)} f(x,y)\,dy - \int_{\alpha}^{\alpha(x)} f(x,y)\,dy$$

$$= g(x, \beta(x)) - g(x, \alpha(x)) \quad \text{连续.}$$

$g(x,t) = \int_{\alpha}^{t} f(x,y)\,dy$

例2. 求 $\lim\limits_{a\to 0} \int_a^{1+a} \frac{1}{1+x^2+a^2}\,dx = \int_0^1 \frac{1}{1+x^2}\,dx = \frac{\pi}{4}$

定理3. 设 $D = [a,b]\times[\alpha,\beta]$. $f(x,u) \in C(D)$.

则 $\varphi(u) = \int_a^b f(x,u)\,dx \in R[\alpha,\beta]$. 且

$$\int_\alpha^\beta \varphi(u)\,du = \int_a^b \left(\int_\alpha^\beta f(x,u)\,du\right) dx$$

$$\parallel$$

$$\int_\alpha^\beta \left(\int_a^b f(x,u)\,dx\right) du$$

证: $\forall\, t \in [\alpha,\beta]$.

$\angle_1(t) = \int_\alpha^t \left(\int_a^b f(x,u)\,dx\right) du$. $\angle_1'(t) = \int_a^b f(x,t)\,dx$.

$\parallel$

$\angle_2(t) = \int_a^b \left(\int_\alpha^t f(x,u)\,du\right) dx$ $\angle_2'(t) = \int_a^b f(x,t)\,dx$.

$\angle_1'(t) = \angle_2'(t)$ $\angle_1(\alpha) = 0 = \angle_2(\alpha)$

令 $g(x,t) = \int_\alpha^t f(x,u)\,du \in C(D)$

$g_t'(x,t) = f(x,t) \in C(D)$

例3. 计算 $\int_0^1 \frac{x^b - x^a}{\ln x}\,dx$. $(b>0,\ a>0)$

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$$\lim_{x\to 0^+} \frac{x^b - x^a}{\ln x} = 0.$$

$$f(x) = \begin{cases} \frac{x^b - x^a}{\ln x} & x\in(0,1] \\ 0 & x=0 \end{cases} \qquad f(x)\in C[0,1].$$

$$F(x,y)=0 \qquad F(x,y)\in C^k.$$

$$\Downarrow$$

$$y=f(x) \in C^k.$$

$$\int_0^1 \frac{x^b - x^a}{\ln x} dx = \int_0^1 \left(\int_a^b x^u du\right) dx = \int_a^b \left(\int_0^1 x^u dx\right) du = \int_a^b \left(\frac{x^{u+1}}{u+1}\Big|_0^1\right) du$$

$$\frac{x^b - x^a}{\ln x} = \frac{x^u}{\ln x}\Big|_a^b = \int_a^b x^u du \qquad (设 a<b)$$

$$= \int_a^b \frac{1}{u+1} du = \ln\frac{b+1}{a+1}$$

$$f(x,u) = x^u \in C(D). \qquad D=[0,1]\times[a,b]$$

例4. 计算 $I(a) = \int_0^\pi \ln(1+a\cos x) dx$. $(|a|<1)$

解: $f(x,a) = \ln(1+a\cos x) \in C(D)$.

$\forall a\in(-1,1)$. $\exists [\alpha,\beta]\subset(-1,1)$ s.t. $a\in[\alpha,\beta]$.

$D=[0,\pi]\times[\alpha,\beta]$.

$$f_a'(x,a) = \frac{\cos x}{1+a\cos x} \in C(D).$$

$a\neq 0$.

$$I'(a) = \frac{1}{a}\int_0^\pi \frac{1+a\cos x - 1}{1+a\cos x} dx = \frac{\pi}{a} - \frac{1}{a}\int_0^\pi \frac{1}{1+a\cos x} dx$$

令 $t=\tan\frac{x}{2}$ $\qquad \cos x = \frac{\cos^2\frac{x}{2} - \sin^2\frac{x}{2}}{\cos^2\frac{x}{2} + \sin^2\frac{x}{2}} = \frac{1-t^2}{1+t^2}$

$x = 2\arctan t$

$$\int_0^\pi \frac{1}{1+a\cos x} dx = \int_0^{+\infty} \frac{1}{1+a\cdot\frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$\int_0 \quad 1+a\cdot\frac{1-t^2}{1+t^2}$$

$$\frac{1+\cancel{t^2}}{\underbrace{1+t^2+a(1-t^2)}_{1+a+(1-a)t^2}}\cdot\frac{2}{\cancel{1+t^2}}$$

$$=\int_0^{+\infty}\frac{2}{1+a+(1-a)t^2}\,dt$$

$$=\frac{\sqrt{1+a}}{\sqrt{1-a}}\cdot\frac{2}{1+a}\int_0^{+\infty}\frac{1}{1+\left(\sqrt{\frac{1-a}{1+a}}\,t\right)^2}\,d\sqrt{\frac{1-a}{1+a}}\,t$$

$$=\frac{2}{\sqrt{1-a^2}}\arctan\left(\sqrt{\frac{1-a}{1+a}}\,t\right)\Bigg|_0^{+\infty}=\frac{\pi}{\sqrt{1-a^2}}.$$

$$\therefore I'(a)=\frac{\pi}{a}-\frac{\pi}{a\sqrt{1-a^2}}$$

$$\therefore I(a)=\pi\ln\left(1+\sqrt{1-a^2}\right)+C.$$

$$\because I(0)=0. \qquad \therefore C=-\pi\ln 2.$$

$$\therefore I(a)=\pi\ln\left(1+\sqrt{1-a^2}\right)-\pi\ln 2.$$

$$\int_a^{+\infty}f(x)\,dx$$

$$\downarrow$$

$$\varphi(u)=\int_a^{+\infty}f(x,u)\,dx$$